

Victoria Knapp Perez

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Summary

Physics Ph.D. candidate with a strong background in machine learning, reinforcement learning, and data-driven modeling, with research experience in theoretical particle physics. Experienced in AI methods such as reinforcement learning (OpenAI Gym), deep learning (PyTorch), and high-performance computing, with an emphasis on automated scientific model discovery and symbolic reasoning. Seeking to apply my AI expertise and problem-solving skills to real-world, impact-driven challenges.

Education

- **University of California, Irvine, USA**, Ph.D. Physics (GPA=4.00) (Expected) June 2026
- **University of California, Irvine, USA**, M.S. Physics (GPA=4.00) 2023
- **National Autonomous University of Mexico, Mexico**, B.Sc. Physics (GPA=9.88/10), Cum Laude 2020

Skills

- Programming & Platforms: Python (7+ years), Java (1+ year), Bash scripting (5+ years), LaTeX (9+ years), Slurm (2+ years), CUDA, and High-Performance Computing (HPC) environments.
- Frameworks and Libraries: PyTorch, FastAI, Stable Baselines, Hugging Face Transformers, scikit-learn, OpenAI Gym, Pandas, NumPy, Matplotlib.
- Machine Learning & Data Science: Reinforcement learning (PPO), deep learning (CNNs, RNNs, transformers, GNNs), attention and sequence-to-sequence architectures, and fine-tuning of pretrained models such as BERT and GPT.
- Development Tools & Version Control: Git, Jupyter Notebook, Linux shell scripting.
- Certifications: [Coursera Java \(2020\)](#), [Deep Learning \(2025\)](#).

Work Experience

University of California: Irvine, USA 2021-Present

Research Assistant & Teaching Assistant

- Engineered end-to-end ML and RL [pipelines](#) to automate physics model construction, integrating PPO agents (Stable Baselines) with HPC training environments (Python, Mathematica, Slurm).
- Published peer-reviewed [papers](#) on theoretical particle physics, cosmology and on [integrating ML methods into physics model discovery](#). Collaborated with cross-disciplinary teams on ML pipeline design.
- Delivered instruction and mentorship across 10+ courses from high-school through PhD level, developing strong communication and leadership skills while simplifying complex topics for diverse audiences.

National Autonomous University of Mexico: Mexico City, Mexico 2020-2021

Research Assistant & Teaching Assistant

- Built Python and Mathematica workflows to simulate and analyze physical models, producing efficient automation for large numerical experiments.
- Delivered instruction across five undergraduate level courses.

Selected projects

AI-assisted Neutrino Flavor Theory Design - University of California: Irvine 2025

- Developed the [AMBer](#) reinforcement-learning framework (PPO, Stable Baselines) and integrated it into an end-to-end pipeline to automate neutrino model generation and parameter optimization. Engineered RL workflows and evaluation modules, enabling large-scale exploration and reducing manual analysis time substantially. Achieved >4× higher model-discovery rates compared to random-walk baselines.
- Built and maintained [FlavorBuilder](#), a Python package for automated model construction and optimization, achieving >5× faster evaluations than previous physics tools.
- Tech stack: Python, PyTorch, Stable Baselines, OpenAI Gym, reinforcement learning (PPO), data analysis, high-performance computing (Slurm).

Text Reliability Classification with Fine-Tuned LLMs - Erdos Institute ML Project 2025

- Fine-tuned a LSTM [model](#) for multi-class truthfulness prediction on Kaggle news datasets, benchmarking against a TF-IDF + logistic regression baseline to quantify LSTM advantages.
- Tech stack: Python, PyTorch, Hugging Face Transformers, SpaCy, scikit-learn, Pandas, NumPy.

Selected publications and awards

- Baretz, J. B., Fieg, M., Ganesh, V., Ghosh, A., Knapp-Perez, V., Rudolph, J., & Whiteson, D. (2025). [Towards AI-assisted Neutrino Flavor Theory Design](#). 2025
- Latino Excellence and Achievement Award (LEAD), honored for research excellence and leadership of Hispanic/Latinx students at UCI. 2024
- UC President's Lindau Nobel Laureate Meetings Fellowship, selected to engage with Nobel Laureates and international scientists. 2024
- UCI Department of Physics and Astronomy Outstanding Graduate Student Award, recognizing top first-year graduate student. 2022

Work authorization

- F-1 OPT (STEM eligible); TN-visa eligible. Open to relocation within the U.S. and internationally.

Physics Publications

1. **Knapp-Perez, V.**, Mohlabeng, G., Ratz, M., & Tait, T. M. (2025). [Thermal History of Non-equilibrated Scalars](#).
2. Huang, F., & **Knapp-Pérez, V.** (2025). [Cosmological stasis from field-dependent decay](#). *Journal of High Energy Physics*, 2025(9).
3. Baur, A., Chen, M.-C., **Knapp-Pérez, V.**, & Ramos-Sánchez, S. (2024). [Modular flavored dark matter](#). *Journal of High Energy Physics*, 2024(12).
4. **Knapp-Pérez, V.**, Liu, X.-G., Nilles, H. P., Ramos-Sánchez, S., & Ratz, M. (2023). [Matter matters in moduli fixing and modular flavor symmetries](#). *Physics Letters. Part B*, 844(138106), 138106.
5. Chen, M.-C., **Knapp-Pérez, V.**, Ramos-Hamud, M., Ramos-Sánchez, S., Ratz, M., & Shukla, S. (2022). [Quasi-eclectic modular flavor symmetries](#). *Physics Letters. Part B*, 824(136843), 136843.
6. Almumin, Y., Chen, M.-C., **Knapp-Pérez, V.**, Ramos-Sánchez, S., Ratz, M., & Shukla, S. (2021). [Metaplectic flavor symmetries from magnetized tori](#). *Journal of High Energy Physics*, 2021(5).
7. Ayala, A., Hernández-Ortiz, S., Hernández, L. A., **Knapp-Pérez, V.**, & Zamora, R. (2020). [Fluctuating temperature and baryon chemical potential in heavy-ion collisions and the position of the critical end point in the effective QCD phase diagram](#). *Physical Review. D*. (2016), 101(7).